Concept Note

Var-Rho Statistics Certification of SiPM Photo-Detector Screening: a Pre-Specified Comparison against the DarkSide-20k Acceptance Criteria

Nasir Khan^1

^1 Independent Researcher — Foundations of Physics & Cosmology,
Wana, South Waziristan, Khyber Pakhtunkhwa, Pakistan
Email: Nasirk684@gmail.com · ORCID: 0009-0002-6006-0796

Document type: concept note (pre-pre-registration). No data are reported.
Archived as: https://doi.org/10.5281/zenodo.23504442 (cite this record)
Lineage: QCID-Sigma v5.4 → muon pre-

registration → propulsion-EMI protocol → swarm veto protocol (10.5281/zenodo.23431140) → swarm validation (10.5281/zenodo.23433610) → front-consistency veto (10.5281/zenodo.23482628) → IceCube DOM study (10.5281/zenodo.23494707) → xenon TPC note (10.5281/zenodo.23499555) → this note.

Keywords: SiPM; quality assurance; photon detector; dark matter;
correlated noise; simulation; pre-registration; Var-Rho statistics
certification; var-rho; statistics-certification

1. Why DarkSide-20k, and why now

DarkSide-20k is in production. The collaboration has published the shape of its photo-detector quality regime: roughly a quarter-million silicon photomultiplier components screened into 648 photo-detector units, with explicit acceptance cuts — per-tile signal-to-noise above 8, pulse-count-rate tails cut at 1.8 Hz/mm², breakdown voltage within ±1 V, a production yield of 87% — and, in companion publications, measured correlated-noise parameters of the sensors themselves (k_dup = 0.445 ± 0.006 in the Proto-0 array; cryogenic characterization reporting correlated delayed avalanches below 0.06 per pulse in a 5 µs window) [1–4]. This note makes one observation about that regime: it is threshold-based, and it is large. Every cut in it is a number chosen by

the collaboration; none of it ships with certified statistics — a locked,
published, reproducible criterion set with Wilson intervals on every reported
rate, against which the kept population, not just the rejected one, is
statistically audited. The DarkSide-20k screening program is, at its scale,
exactly the operation that Var-Rho statistics certification was designed for,
and its published criteria make the comparison unusually clean: the
collaboration's own numbers are the comparator.

2. What is proposed

A pre-specified simulation study that takes the published DarkSide-20k
parameterizations — tile SNR and PCR

distributions, breakdown spreads, correlated-avalanche parameters — generates screening populations, and evaluates them twice: once under the published acceptance criteria, once under the Var-Rho statistics certification protocol (the locked two-stage gate at Var > 1000, ρ > 0.8, hold-off re-derived per rate regime, with the pulse / interval / rate three-layer evaluation and falsifier discipline established in the IceCube study). The deliverable is not a claim of superiority. It is a side-by-side on identical simulated populations: where the two regimes agree, the study reports agreement as evidence that the Var-Rho criteria are consistent with the collaboration's own engineering judgment;

where they disagree, the study characterizes which tiles each regime keeps or rejects and quantifies the difference with intervals. Either outcome is publishable, and the criteria for both are locked before the first simulated tile exists.

3. Data reality, stated plainly

No DarkSide-20k raw data are public, and this note claims none. Everything Phase 1 uses is published: acceptance thresholds, measured parameter distributions, and correlated-noise measurements, cited below. Phase 2 — application to real screening streams — is conditional on engagement with the collaboration's QA effort, and is not promised

here. What Phase 1 can settle
from a desk, on published numbers, in the
collaboration's own units, is
whether a locked, certifiable screening criterion
set agrees with the one a
major collaboration arrived at by engineering
judgment. That question is
worth a paper either way.

4. Mapping

Table 1. From the program baseline to the DarkSide-20k screening setting.

Baseline (QCID / IceCube / xenon) DarkSide-20k setting
Single-photon pulses (SiPM/SiPM-like) SiPM tile / PDU pulses, cryogenic operation
Correlated noise component k_dup, correlated

delayed avalanches (published)
Amplitude/discriminator comparator SNR > 8,
PCR tail cut, breakdown window (published cuts)
Poisson rate-honesty, dead-time-corrected
Certified statistics of the kept population
Falsifier families S / N / X Falsifiers D1–D6
(Table 2)
Seed-pooled Run-All notebook Same discipline

5. Falsifier framework

Proposed defaults; numeric thresholds are
locked in the planned
pre-registration before any simulation runs.

Table 2. Proposed falsifiers.

ID Claim protected Proposed criterion (fixed at
pre-registration)

D1 Correlated-noise sensitivity ≥ 95% identification of tiles whose correlated-avalanche parameters exceed the published Proto-0 / cryogenic values, at locked thresholds
D2 Signal efficiency ≥ 90% retention of tiles that pass the published acceptance criteria (agreement floor)
D3 Rate-honesty Certified statistics hold on the kept population at all scanned rates; Fano within [0.9, 1.1] at the nominal regime
D4 Threshold portability Locked operating point unchanged across simulated batch-to-batch spreads (published yields and breakdown distributions)
D5 No bias Efficiency flat across breakdown-voltage and gain bins within tolerance
D6 Head-to-head vs published criteria On identical simulated populations: keep/reject agreement with the published cuts quantified (agreement fraction with intervals); claims

beyond agreement bounded by the quantified difference, locked in advance

6. Status and next step

This note fixes the question and the falsifier framework in advance, in the
manner of the program's records before it. The sequel is a locked
pre-registration — D6's agreement criteria frozen with everything else — then
the Phase-1 simulation on published parameters, archived with results and
figures. The conversation this work is for begins with the locked protocol
and the result in hand, offered to the screening effort as a comparison it
does not have to run itself to read.

Declarations

Funding: None (independent research).
Competing interests: None.
Data availability: No data accompany this note.
Author contributions:
N.K. performed the reconnaissance and wrote this note.
Correspondence: Nasirk684@gmail.com

References

1. DarkSide Collaboration, "Construction and characterisation of the DarkSide-20k veto SiPM tiles" (2026) — SNR > 8, PCR tail at 1.8 Hz/mm², breakdown ±1 V, 87% yield. [exact citation at pre-registration]

2. DarkSide Collaboration, Proto-0 array paper

(2026), arXiv — k_dup = 0.445 ± 0.006. [exact citation at pre-registration]

3. DarkSide Collaboration, "Characterization of the Hamamatsu VUV4 SiPMs at cryogenic temperatures" (Oct 2026), arXiv:2610.18223 — correlated delayed avalanches < 0.06 per pulse, 5 μs window.

4. DarkSide Collaboration, photo-detector production database (PoS). [exact citation at pre-registration]

5. N. Khan, QCID-Sigma v5.4, Zenodo 10.5281/zenodo.23378709 — hardware-verified baseline.

6. N. Khan, muon pre-registration v1.1, Zenodo 10.5281/zenodo.23405991.
7. N. Khan, swarm veto protocol v1.0, Zenodo 10.5281/zenodo.23431140.

8. N. Khan, swarm veto computational validation, Zenodo 10.5281/zenodo.23433610.
9. N. Khan, front-consistency veto record, Zenodo 10.5281/zenodo.23482628.
10. N. Khan, IceCube DOM noise record, Zenodo 10.5281/zenodo.23494707 — hold-off re-derivation and three-layer precedent.
11. N. Khan, xenon TPC note, Zenodo 10.5281/zenodo.23499555.